

Fragment Parsing

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1 Background

1.1 Context-Free Grammars (copied from satta1994)

A *context-free grammar* (CFG) is a four-tuple $G = (N, \Sigma, P, S)$, where N is a finite set of nonterminal symbols, Σ is a finite set of terminal symbols with $N \cap \Sigma = \emptyset$, P is a finite set of productions, and $S \in N$ is the start symbol.

Productions in P have the form $D_r \rightarrow Z_{r,1}Z_{r,2}\dots Z_{r,\pi_r}$, where $D_r \in N$ and $Z_{r,j} \in N \cup \Sigma$, $\pi_r \geq 0$. A production is *null* if $\pi_r = 0$ and a *chain* production if $\pi_r = 1$ and $Z_{r,1} \in N$.

The language derived by G is $L(G) = \{w \mid S \xrightarrow{*} w, w \in \Sigma^*\}$.

1.2 Decomposition

Let ω be an input string that can be decomposed as

$$\omega = \alpha \cdot \beta \cdot \gamma$$

where $\alpha, \gamma \in \Sigma^*$ and $\beta \in \Sigma \cup N$.

There exists a derivation

$$\alpha\beta\gamma \Rightarrow \alpha A \gamma$$

or generally,

$$\alpha\beta\gamma \Rightarrow [\alpha/x]A[\gamma/y]$$

where $x, y \in \Sigma^* \cup N^* \cup \{\varepsilon\}$. The notation $[\alpha/x]$ denotes a correspondence between the input flank α and the production symbol x — specifically, $x \xrightarrow{*} \alpha$ (x derives α). The precise treatment of the recursive case is given in Section 3.

The possible values for the input flanks (α, γ) and production flanks (x, y) are:

I — input flanks (α, γ)	II — production flanks (x, y)
(1) $\alpha = \varepsilon, \gamma = \varepsilon$	(1) $x = \varepsilon, y = \varepsilon$
(2) $\alpha = \varepsilon, \gamma \in \Sigma^+$	(2) $x = \varepsilon$
(3) $\alpha \in \Sigma^+, \gamma = \varepsilon$	(3) $y = \varepsilon$
(4) $\alpha, \gamma \in \Sigma^+$	(4) $x, y \neq \varepsilon$

1.3 Context Reduction Matrix

Mixing and matching cases from I and II gives all possible reduction scenarios for β . Given a production $A \rightarrow x\beta y$ and input fragment $\alpha\beta\gamma$, the grammar defines what flanks are expected. A flank is a *gap* if the production requires it but the input does not supply it.

Two notations are used in the derivations below:

- (x) — x is a production symbol required by the grammar but absent from the input (a gap)
- $[\alpha/x]$ — α is present in the input and corresponds to production symbol x

Cases 1.x — $\alpha = \varepsilon, \gamma = \varepsilon$:

(1.1)	$\beta \rightarrow A$
(1.2)	$\beta \rightarrow \beta(y) \rightarrow A$
(1.3)	$\beta \rightarrow (x)\beta \rightarrow A$
(1.4)	$\beta \rightarrow (x)\beta(y) \rightarrow A$

Cases 2.x — $\alpha = \varepsilon, \gamma$ present:

(2.1)	$\beta\gamma \rightarrow A\gamma$
(2.2)	$\beta\gamma \rightarrow \beta[\gamma/y] \rightarrow A\gamma'$
(2.3)	$\beta\gamma \rightarrow (x)\beta\gamma \rightarrow A\gamma$
(2.4)	$\beta\gamma \rightarrow (x)\beta[\gamma/y] \rightarrow A\gamma'$

Cases 3.x — α present, $\gamma = \varepsilon$:

(3.1)	$\alpha\beta \rightarrow \alpha A$
(3.2)	$\alpha\beta \rightarrow \alpha\beta(y) \rightarrow \alpha A$
(3.3)	$\alpha\beta \rightarrow [\alpha/x]\beta \rightarrow \alpha' A$
(3.4)	$\alpha\beta \rightarrow [\alpha/x]\beta(y) \rightarrow \alpha' A$

Cases 4.x — α present, γ present:

(4.1)	$\alpha\beta\gamma \rightarrow \alpha A\gamma$
(4.2)	$\alpha\beta\gamma \rightarrow \alpha\beta[\gamma/y] \rightarrow \alpha A\gamma'$
(4.3)	$\alpha\beta\gamma \rightarrow [\alpha/x]\beta\gamma \rightarrow \alpha' A\gamma$
(4.4)	$\alpha\beta\gamma \rightarrow [\alpha/x]\beta[\gamma/y] \rightarrow \alpha' A\gamma'$

1.3.1 Note on Null Productions

We assume β is never nullable. For nullable flanks (x or y), an implementation may precompute the nullable set $\mathcal{N}_G = \{A \in N \mid A \xrightarrow{*} \varepsilon\}$ and add skip-rules: for any case where a flank symbol is in $\mathcal{N}_G$, add the corresponding case with that flank dropped. For example, if $x \in \mathcal{N}_G$, case 4.3 ($[\alpha/x]\beta\gamma \rightarrow \alpha' A\gamma$) admits a skip to case 2.1 ($\beta\gamma \rightarrow A\gamma$) directly. Epsilon productions need not fire at runtime.

1.4 Recurrence

Let $R(\omega) \subseteq N$ denote the set of nonterminals that span ω . The recurrence is defined as follows:

Base case:

$$R(\beta) = \{A \in N \mid A \rightarrow x\beta y \in P, \\ x, y \in N \cup \Sigma \cup \{\varepsilon\}\}$$

where x and y are either matched against input flanks or treated as gaps per the CR matrix. This is a fixed-point computation over a finite set — cycles in the grammar do not cause non-termination.

Inductive step: For $\omega = \alpha\beta\gamma$, apply the CR matrix to obtain $\alpha' A\gamma'$ for each $A \in R(\beta)$:

$$R(\omega) = \bigcup_{\{A \in R(\beta)\}} R(\alpha' A\gamma')$$

where $|\alpha' A\gamma'| < |\omega|$, guaranteeing termination.

A *final* nonterminal F is any $A \in R(\omega)$ such that A spans the entire input — i.e. no unreduced symbols remain. F need not be the start symbol S ; any nonterminal covering ω constitutes a valid parse result.